

Heart Disease Risk Prediction - MLOps Assignment Report

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Area	Implementation
Dataset	UCI Heart Disease using ucimlrepo id=45
Models	Logistic Regression and Random Forest
Tracking	MLflow with SQLite
API	FastAPI /predict, /health, /metrics
Deployment	Docker and Kubernetes manifests
Monitoring	API logs and Prometheus metrics

1. Project Overview

I developed an end-to-end MLOps workflow for binary heart disease risk prediction. The project covers data acquisition, EDA, preprocessing, model training, tracking, packaging, API serving, Docker, CI/CD, Kubernetes files, and basic monitoring.

2. Dataset Acquisition

The dataset is fetched with ucimlrepo dataset id 45. The target is converted into a binary label where zero means no heart disease and one means disease present.

3. EDA

The EDA script creates class balance, histogram, correlation heatmap, and missing value plots. I will add screenshots from my own run before submission.

4. Preprocessing

I used a ColumnTransformer with numeric imputation/scaling and categorical imputation/encoding. It is saved inside the final pipeline.

5. Model Development

I trained Logistic Regression and Random Forest models using GridSearchCV with stratified 5-fold cross-validation and ROC-AUC scoring.

6. MLflow Tracking

MLflow logs parameters, metrics, ROC curves, confusion matrices, and joblib model artifacts. SQLite is used as the tracking backend.

7. API and Packaging

FastAPI serves the saved model through /predict and exposes /health and /metrics endpoints.

8. CI/CD, Docker, Deployment

GitHub Actions, Dockerfile, and Kubernetes YAML files are included. I need to add screenshots after running these on my system.

9. Monitoring

The API logs requests and exposes Prometheus-compatible metrics. Starter Prometheus and Grafana files are included.

10. Final Items

Before final submission, I will add screenshots, GitHub link, deployment/access instructions, and a video link.

Main commands

```
python -m venv mlops
.\mlops\Scripts\Activate.ps1
pip install --prefer-binary -r requirements.txt
python src\data_acquisition.py
python src\eda.py
python src\train.py
pytest -q
uvicorn api.main:app --reload --host 0.0.0.0 --port 8000
mlflow ui --backend-store-uri sqlite:///mlflow.db
```